

Unbiased Galaxy Clustering With Fewer Galaxies

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ABSTRACT

Galaxy clustering statistics constrain the physics of galaxy formation, yet remain underutilised in model calibration due to the computational cost of measuring two-point correlation functions for cosmological simulations. When using Semi-Analytic Models (SAMs), halo catalogues are split into k independent realisations of the full box. Estimating the two-point correlation function (2PCF) on a naively combined subset of $m < k$ partitions, normalised by k/m biases the small-scale clustering signal by a factor of that normalisation, because one-halo and two-halo pairs scale differently with partition fraction. We introduce SCOPE (Sub-volume Correction On Pair-count Estimators), a theoretically exact correction scheme that removes this bias using weights $\alpha = m/k$ and $\beta = m(k-1)/[k(m-1)]$ applied to within-partition and between-partition pair counts respectively. The correction depends only on partition geometry, not on galaxy-formation physics, making it universally applicable across simulations. Applied to GALFORM predictions across three N-body simulations (L800, Millennium I, and Millennium II), SCOPE achieves sub-percent accuracy at one-halo scales ($r < 1 h^{-1}$ Mpc) and 2-3 per cent accuracy at two-halo scales ($5 < r < 30 h^{-1}$ Mpc) using a target partition fraction of ($f = m/k$), $f_{\text{target}} = 6\%$. The method is robust across galaxy-formation physics (validated on 1c16 and gp14 GALFORM variants), cosmic time ($z = 0$ to $z = 1.5$) and galaxy selection (stellar mass 10^9 to $10^{11} M_{\odot} h^{-1}$). Computationally, SCOPE enables significant savings in model calibration: by running the SAM on only 6% of the halo merger trees (rather than the full simulation), it achieves 2-3 per cent large-scale clustering accuracy. This efficiency enables rapid exploration of parameter space for likelihood-free inference and establishes that clustering statistics can now be included as a calibration tool for galaxy-formation models.

Key words: methods:numerical – methods:statistical – galaxies:clustering – galaxies:formation – dark matter

1 INTRODUCTION

Historically, semi-analytic galaxy formation models (SAMs) have relied on one-point statistics (commonly the luminosity function and stellar mass functions) as their primary constraints for model calibration (Baugh 2006). By requiring models to reproduce these statistics to match observations across different redshifts, SAMs have been able to provide insights into the physical processes that govern galaxy formation and evolution. However, relying on one-point statistics in this way has its limitations. Different models are capable of reproducing these same one-point statistics, making it difficult to distinguish validity between them. Further to this, one-point statistics are not sensitive to the spatial distribution of galaxies and thus do not capture the full complexity of galaxy clustering. This can lead to degeneracies in model parameters, where different combinations of parameters can produce similar one-point statistics but different clustering properties.

To approach this problem, we require a method to quantify the clustering of galaxies to be sensitive to their spatial distribution. For this, a common approach is to use the two-point correlation func-

tion (2PCF), which measures the excess probability of finding pairs of galaxies at a given separation compared to a random distribution. By comparing the 2PCF of simulated galaxies to that of observed galaxies, we can constrain the parameters of galaxy formation models and gain insights into the physical processes that govern galaxy clustering.

Because galaxies reside within their dark matter halos, their clustering acts as a tracer of the underlying dark matter distribution (Cooray & Sheth 2002). This link - the galaxy-halo connection, shapes galaxy formation and evolution. By studying the clustering of galaxies, we gain insights into the properties of dark matter halos, such as their mass and spatial distribution, and how they evolve over time. This information is essential for understanding the formation and evolution of galaxies, as well as for constraining cosmological models.

Despite the potential constraining power of galaxy clustering, it has rarely been utilised in the calibration of galaxy formation models. One of the main barriers to entry is the computational cost of calculating the 2PCF for large cosmological simulations. Modern cosmological simulations involve many millions of galaxies and the 2PCF requires counting pairs of galaxies at different separations, which scales $O(\text{galaxy count}^2)$ - computationally expensive.

Existing approaches to reduce this cost fall into two categories. The first, Jackknife resampling and bootstrap methods (Efron & Tibshirani 1993) provide unbiased covariance estimates but require

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computing the 2PCF independently for each resample - multiplying the computational cost by the number of resamples. Alternatively, practitioners use sparse subsamples of the full simulation, but naively combining them causes differences in the small- and large-scale regimes - causing a bias that needs to be treated and makes the intermediate-scale turnover especially difficult. The latter approach is computationally attractive but has lacked a principled correction scheme.

As we enter the era of large-scale galaxy surveys - such as the Dark Energy Spectroscopic Instrument (DESI) (DESI Collaboration 2016) and Euclid (Laureijs et al. 2011) - observational uncertainties on clustering statistics are shrinking to unprecedented levels. This has made the constraining power of clustering increasingly important to include clustering in the calibration of galaxy formation models.

Furthermore, addressing modern tensions in cosmology - such as the Hubble tension (Riess et al. 2022) and the S8 tension (Planck Collaboration 2020) - requires a detailed understanding of the clustering of galaxies and its connection to the underlying dark matter distribution. This requires moving beyond empirically calibrated Halo Occupation Distribution (HOD) models to physical galaxy formation models.

Likelihood-free inference methods, such as simulation-based inference (SBI) and neural density estimation (NDE), could potentially revolutionise galaxy formation model calibration, however, these methods require evaluating the forward model and computing its summary statistics thousands of times or more. While SAMs are inherently faster than hydrodynamical simulations, applying high-resolution SAMs with Gpc/h box sizes and subsequently computing the 2PCF for millions of galaxies for each calibration step is still computationally prohibitive.

Alternative approaches often involve either relying on weak convergence criteria - using small boxes (introducing sample variance and missing large-scale information) or must restrict the dimensionality of the parameter space they can explore; neither represents ideal circumstances for cosmological inference.

In this paper, we demonstrate that this computational bottleneck can be bypassed while maintaining accuracy. We introduce a new method for unbiased estimation of the 2PCF using a fraction of the galaxies in the simulation - SCOPE (Sub-volume Correction On Pair-count Estimators). This method preserves the baryonic physics embedded within the SAM while accurately estimating the full-box 2PCF at all scales. This allows the use of the 2PCF in galaxy formation model calibration and inference. In Section 5 we demonstrate that when applied to GALFORM, SCOPE achieves 2–3 per cent large-scale accuracy on only 6 per cent of the halo merger trees - a ~ 16 -fold reduction in SAM cost.

2 GALAXY FORMATION MODEL

To validate that SCOPE works across different galaxy-formation physics, we require a semi-analytic model that produces assembly-history-dependent clustering properties. We use the GALFORM model (Cole et al. 2000; Bower et al. 2006; Lacey et al. 2016), which evolves baryonic components through cosmic time within dark matter halo merger trees extracted from N -body simulations. This physically motivated approach is ideal for our validation because the clustering statistics it predicts depend on the full assembly history of halos, not just their instantaneous mass (such as in an empirical model).

2.1 Halo Occupation

The HOD describes how galaxies populate dark matter halos and is the immediate driver of clustering statistics. In GALFORM, the HOD emerges as a natural consequence of the full assembly history of each halo: the model evolves baryonic components through cosmic time using physically motivated prescriptions for gas cooling, star formation, stellar feedback, AGN growth and galaxy mergers. Crucially, two halos with identical present-day mass can host qualitatively different galaxy populations if their formation times, merger histories, or internal structures differ. This assembly-history dependence generates clustering signatures that are absent in empirical models parameterising occupation as functions of halo mass alone.

2.2 Key processes and physics parameters

The clustering statistics predicted by GALFORM depend on physical processes that shape the halo occupation distribution. Galaxy mergers occur on dynamical-friction timescales that control the satellite fraction at fixed halo mass, directly setting the small-scale clustering amplitude. Disk instabilities and bar formation provide an alternative channel for galaxy evolution and quenching, affecting the distribution of intermediate-mass galaxies ($M_* \sim 10^{10}$) and their clustering (where M_* is the total stellar mass of a galaxy). Supernova and AGN feedback regulate the star formation histories of central and satellite galaxies, controlling both the abundance of galaxies and their spatial distribution within halos. Because these processes are calibrated against observational constraints and have tunable parameters, different GALFORM variants can produce significantly different clustering predictions.

2.3 Validation Design: Models, Simulations, and Samples

2.3.1 Model variants

If SCOPE succeeds across models with qualitatively different halo occupation distributions (HODs), it demonstrates the correction factors $\alpha = m/k$ and $\beta = m(k-1)/[k(m-1)]$ (Section 3) depend only on the structure of halo partitions, not on the galaxy-formation physics that generates clustering. We use two GALFORM variants. The fiducial model, 1c16 (Lacey et al. 2016), is calibrated to the observed stellar mass function at $z = 0$. The alternative, gp14 (González-Pérez et al. 2014), uses different supernova and AGN feedback prescriptions and produces a different HOD: at fixed stellar-mass threshold, gp14 predicts different satellite-to-central ratios and small-scale clustering amplitudes than 1c16 (Table 1).

2.3.2 Partitioning

Each N -body simulation's halo merger trees are split into k subsets at the simulation level. The full set of merger trees is distributed across k identically-sized tree files at redshift zero and GALFORM runs independently on each. Selecting and combining $m \leq k$ partitions then yields a density-reduced realisation of the galaxy distribution, which SCOPE corrects back to the full-box prediction.

These partitions are not spatial divisions of the box. Galaxies from each partition occupy the entire volume; the distinction between partitions is in merger-tree membership alone. Each partition is therefore an independent, lower-density realisation of the full-volume galaxy distribution. Figure 1 contrasts this with the common, spatial sub-volume division.

Partitioning at the merger-tree level has a clean consequence for

Parameter	Description	1c16	gp14
<i>Supernova feedback</i>			
V_{hot}	Ejection velocity threshold (km s ⁻¹)	320	380
α_{hot}	Mass-loading power-law slope	3.4	3.2
<i>AGN feedback</i>			
ϵ_{AGN}	Eddington accretion-rate fraction	0.010	0.040
α_{cool}	Cold/hot accretion threshold	0.80	0.61
<i>Satellite evolution</i>			
τ_0	Dynamical-friction timescale normalisation	1.0	1.5
α_{reheat}	Gas reincorporation factor	1.00	1.26
Starvation	Satellite cooling suppressed	Yes	No
<i>Star formation and stellar populations</i>			
ν_{SF}	H ₂ depletion rate (Gyr ⁻¹)	0.74	0.50
f_{stab}	Disk stability threshold	0.90	0.80
IMF	Stellar initial mass function	Bimodal ^a	Chabrier

^a Kennicutt (1983) for quiescent star formation; top-heavy ($x = 1$) for burst-mode star formation.

Table 1. Key GALFORM parameters for the two model variants used in this work. Parameters not listed are identical between 1c16 and gp14. The supernova feedback velocity V_{hot} applies to both disc-mode and burst-mode star formation. ϵ_{AGN} is the Eddington accretion-rate fraction determining radio-mode AGN heating. τ_0 rescales the dynamical-friction merger timescale. α_{reheat} sets the reincorporation timescale for feedback-ejected gas. The IMF affects stellar mass-to-light ratios and the calibration target.

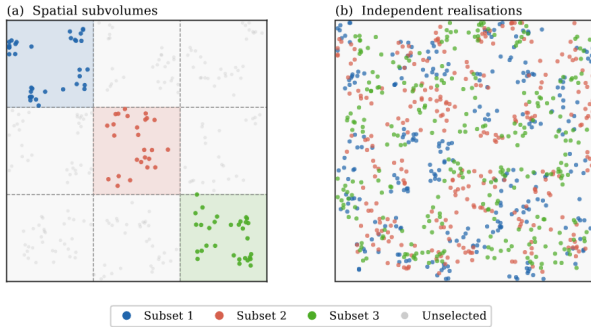


Figure 1. Partition architecture. Left: spatial sub-volume division, with galaxies grouped by location. Right: the merger-tree partitioning used here, with partition labels assigned by merger-tree membership rather than position. Partition labels are spread throughout the full volume, preserving the spatial distribution while cleanly separating one-halo and two-halo pair counts.

pair counts. No halo is split across partitions, so every one-halo pair (two galaxies in the same halo) belongs to a single partition. Two-halo pairs (galaxies in different halos) can belong to the same partition or to different ones. Selecting $m < k$ partitions therefore scales one-halo pair counts as m/k and two-halo pair counts as m^2/k^2 . This separation is what makes the correction weights $\alpha = m/k$ and $\beta = m(k-1)/[k(m-1)]$ possible (Section 3).

Simulation	L	k	Cosmology	Reference
L800	542.2	1024	Planck 2013	Baugh et al. (2019)
Millennium I	500	64	WMAP1	Springel et al. (2005)
Millennium II	100	64	WMAP1	Boylan-Kolchin et al. (2009)

Table 2. Summary of N-body simulations used. L is the box size in units of h^{-1} Mpc, k is the number of halo partitions and the cosmology column lists the underlying cosmological parameters. All simulations are processed with GALFORM.

2.3.3 Simulations

We use three dark matter, N-body simulations, processed with GALFORM. Each simulation's halo population is partitioned into k subsets as described above. Table 2 summarises their properties.

- **L800:** The large $k = 1024$ allows convergence to be tested over more m values.
- **Millennium I:** Compares 1c16 and gp14 on the same volume.
- **Millennium II:** Higher resolution and smaller volume; extends the halo mass range probed.

2.3.4 Galaxy sample definitions

We use four galaxy samples, applied uniformly across all simulations and redshifts. The samples span one-halo-dominated to two-halo-dominated clustering:

- All-galaxy catalogue** ($M_* > 0$): $\sim 9,000,000$ galaxies per L800 partition at $z = 0$. All central and satellite galaxies, no mass threshold. Strong one-halo signal.
- $M_* > 10^9 M_\odot h^{-1}$: $\sim 408,000$ galaxies per L800 partition at $z = 0$. One-halo dominated; the regime in which naive partition combination produces the largest small-scale bias ($\sim k/m$).
- $M_* > 10^{10} M_\odot h^{-1}$: $\sim 51,000$ galaxies per L800 partition at $z = 0$. one-halo and two-halo terms comparable.
- $M_* > 10^{11} M_\odot h^{-1}$: ~ 630 galaxies per L800 partition at $z = 0$. Two-halo dominated and shot-noise limited at small scales; tests β in the sparse regime.

At high M_* thresholds the $\alpha = m/k$ correction must suppress a small-scale bias of $\approx k/m$; at low thresholds the $\beta = m(k-1)/[k(m-1)]$ correction must recover the full-box two-halo signal from sparse cross-pairs.

Convergence is measured at $z \approx 1.5, 0.5$ and 0.0 , the range covered by DESI and Euclid and within which nonlinear clustering is well-described by halo models. Behaviour at $z > 2$, where halo assembly is rapid and the nonlinear regime more extreme, is not validated here.

2.3.5 Model assumptions and systematic uncertainties

GALFORM, like other semi-analytic models, carries systematic uncertainties: orphan satellite evolution below resolution limits, approximate environmental processing and calibration of merger timescales and feedback thresholds. These affect small-scale pair counts and the one-halo to two-halo transition and contribute to the ~ 10 -20 per cent clustering amplitude differences between 1c16 and gp14.

3 CLUSTERING FORMALISM AND ESTIMATOR

We derive the estimator for $\xi(r)$ measured from the halo-partitioned catalogues described in Section 2. Because running GALFORM on all k partitions is computationally prohibitive for model calibration, we

seek an estimator that recovers the full-box $\xi(r)$ from only $m < k$ selected partitions.

Galaxy pair counts decompose into one-halo pairs (both galaxies in the same halo) and two-halo pairs (galaxies in different halos). This decomposition is central because these pair types scale differently with density when a subset $m < k$ of partitions is used, as formalised in Section 3.2.

3.1 Baseline full-sample estimator

In each radial bin centred on r ,

$$\hat{\xi}(r) = \frac{DD_{\text{norm}}(r)}{RR_{\text{per}}(r)} - 1, \quad (1)$$

where

$$DD_{\text{norm}}(r) = \frac{DD(r)}{N_D(N_D - 1)} \quad (2)$$

is the normalised pair fraction, where $DD(r)$ is the count of distinct galaxy pairs whose separation falls in the radial bin centred on r , and N_D is the total number of galaxies in the catalogue. The key computational advantage is that $RR_{\text{per}}(r)$, the pair fraction expected for a uniform random distribution in the periodic box (computed analytically from shell volume and box volume), replaces the Monte Carlo random catalogue required by conventional estimators such as Landy-Szalay. This eliminates a major computational cost of traditional 2PCF calculations. The full-box measurement is taken as the reference against which sub-sampled estimates are compared.

3.2 Bias in the combined catalogue

Let DD_{1h} and DD_{2h} denote the absolute number of one-halo and two-halo pairs in the complete, full-box simulation. When we combine m selected partitions, we create a single catalogue at reduced density m/k (lower than the full-box density by factor k/m). The straightforward approach is to compute the two-point correlation function on this combined catalogue and then scale it by the density correction factor k/m to account for the fact that we have fewer galaxies.

However, this naive density correction fails because one-halo and two-halo pairs do not scale identically with the density reduction. To see this, decompose pair counts by partition membership. Let DD_{auto} be pairs within each partition, and DD_{cross} be pairs between different partitions.

One-halo pairs comprise galaxies in the same halo and must therefore reside in the same partition. When m of the k partitions are selected, the number of one-halo pairs scales with the density reduction as m/k . Two-halo pairs comprise galaxies in different halos. If each halo is independently assigned to one of the k partitions uniformly at random, the probability that two distinct halos are both assigned to the same partition among the selected m is m/k^2 , and the probability they are assigned to different partitions is $m(m-1)/k^2$. Thus in expectation:

$$\langle DD_{\text{auto}} \rangle = \frac{m}{k} DD_{1h} + \frac{m}{k^2} DD_{2h}, \quad (3)$$

$$\langle DD_{\text{cross}} \rangle = \frac{m(m-1)}{k^2} DD_{2h}. \quad (4)$$

In aggregate, two-halo pairs scale as $\frac{m}{k^2} + \frac{m(m-1)}{k^2} = \frac{m^2}{k^2}$.

This independence and uniformity assumption is essential: if partition labels were correlated with halo position or mass, the same-partition probability would become scale- or mass-dependent,

adding correction terms to the coefficients above. We return to this constraint in Section 3.4.

A naive global correction by factor k/m correctly recovers the two-halo signal (since the aggregate two-halo scaling m^2/k^2 matches the density factor), but it over-corrects the one-halo signal. It scales the measured one-halo pairs (which scale as m/k) by k/m , creating an inflation of the one-halo signal by a residual factor of k/m . For the canonical case of $k = 1024$ and $m = 8$, the naive density correction thus inflates the one-halo signal by a factor of 128.

3.3 Auto/cross correction

The solution is to apply different correction weights to the auto and cross terms. Since no halo is split across partitions by construction, every one-halo pair must carry the same partition label (both galaxies from the same halo belong to the same partition), while every cross-partition pair is guaranteed to be a two-halo pair. This means we can separate them and reweight them independently.

We seek a corrected pair count of the form

$$DD_{\text{corr}} = \alpha DD_{\text{auto}} + \beta DD_{\text{cross}}, \quad (5)$$

where $DD_{\text{auto}} = \sum_{i=1}^m DD_{ii}$ and $DD_{\text{cross}} = \sum_{i \neq j} DD_{ij}$. We want $\langle DD_{\text{corr}} \rangle$ to equal $\frac{m^2}{k^2} (DD_{1h} + DD_{2h})$, the full-box pair count rescaled by the density factor $(m/k)^2$ that normalisation by N_D^2 introduces.

Using the scaling relations from Section 3.2, we match coefficients. For the one-halo term (which appears only in DD_{auto}):

$$\alpha \frac{m}{k} = \frac{m^2}{k^2} \Rightarrow \alpha = \frac{m}{k}, \quad (6)$$

For the two-halo term (which appears in both auto and cross):

$$\alpha \frac{m}{k^2} + \beta \frac{m(m-1)}{k^2} = \frac{m^2}{k^2} \Rightarrow \beta = \frac{m(k-1)}{k(m-1)}. \quad (7)$$

In the limit $k \gg m \gg 1$, one has $\alpha \rightarrow m/k \ll 1$ and $\beta \rightarrow 1$. In practice, the correction mostly down-weights the auto pairs (which disproportionately contain the one-halo signal), while leaving the cross-pairs nearly unchanged. At the opposite extreme, setting $m = k$ gives $\alpha = \beta = 1$ and recovers the standard full-catalogue pair count (as expected, since there is no density correction needed when we use all partitions). The correction effectively reweights the auto and cross terms to produce the pair-count distribution that would result from the full-box catalogue at full density, even though we are working with the m/k density combined catalogue. The smallest useful configuration is $m = 2$, since this is the minimum value for which cross-pairs exist.

Substituting the corrected pair count into the periodic-box, isotropic estimator gives

$$\hat{\xi}(r) = \frac{DD_{\text{corr, norm}}(r)}{RR_{\text{per}}(r)} - 1, \quad (8)$$

with

$$DD_{\text{corr, norm}}(r) = \frac{DD_{\text{corr}}(r)}{N_D(N_D - 1)}. \quad (9)$$

This is the SCOPE estimator, where N_D is the galaxy count of the m -partition combined catalogue.

The dominant cost remains the pair count on the combined catalogue, which scales as N_D^2 . The per-partition DD_{auto} calculations are cheaper by roughly $1/m^2$ each and are therefore sub-dominant for the configurations considered here.

3.4 Assumptions and validity regime

The correction relies on the following assumptions. First, halos must not be split across partitions: each halo must belong to exactly one partition at the simulation level. Second, catalogue selection must be by partition label rather than by individual galaxies. Third, the underlying N-body dark-matter simulation must use periodic boundary conditions (standard for large-scale cosmological simulations). The SCOPE method then leverages this periodic geometry to compute $RR_{\text{per}}(r)$ without the need for random catalogues.

Redshift-dependent bias: Beyond this geometric scaling, an additional redshift-dependent bias arises from the partition approach. As the merger trees from the dark matter runs are partitioned today (redshift $z = 0$). At earlier epochs ($z > 0$), halos in different partitions have distinct merger histories, introducing a systematic offset in their clustering properties relative to the full-volume case. Section 5.2 confirms that the α/β correction remains robust across the full redshift range tested despite this effect.

Under these conditions, the correction cleanly separates the one-halo and two-halo contributions and removes the small-scale inflation introduced by naive combination of partition catalogues. The method is most powerful when $m \ll k$, where the naive bias is largest; it reduces smoothly to the standard full-sample estimator when all partitions are included ($m = k$).

4 EVALUATION METRICS

4.1 Reference

The reference $\xi_{\text{ref}}(r)$ is the two-point correlation function computed on the merged catalogue of all k partitions, using the periodic-box estimator of Section 3. The validation matrix - three N-body simulations, two GALFORM variants, four stellar-mass selections, three redshifts - is set out in Section 2 and applied identically to the reference and to every SCOPE partition selection.

4.2 Partition selection protocol

A fixed ordering of partitions is used for every m : each step adds the next partition in the ordering, so every combination is a strict superset of the previous one. This isolates the effect of increasing m from variation in the underlying halo subset. Convergence under randomised partition selection is examined separately in Section 5.6.

4.3 Separation bins

Pair counts are tabulated in 30 logarithmically spaced bins covering $0.01 \leq r \leq 271 h^{-1} \text{ Mpc}$, with a bin width of 0.148 dex. Convergence metrics use the first 27 bins ($0.01 \leq r \leq 100 h^{-1} \text{ Mpc}$); the three outermost bins approach the box scale and are excluded.

4.4 Accuracy metrics

The primary metric is the median absolute log-ratio,

$$\mathcal{L}(m) = \text{median}_{r \in \mathcal{R}} \left| \log_{10} \frac{\xi(r)}{\xi_{\text{ref}}(r)} \right|, \quad (10)$$

where $\mathcal{R}$ denotes the bins in which both ξ and ξ_{ref} are strictly positive. The log-ratio is symmetric on the log scale, so a factor-of-two over- and under-estimate carry equal weight, and it maps directly to the chi-squared contribution of a log-space galaxy-clustering likelihood, $\chi^2 = \sum_r (\log \xi_{\text{model}} - \log \xi_{\text{obs}})^2 / \sigma_r^2$. The median is preferred over

the mean because $\xi(r)$ spans five orders of magnitude across $\mathcal{R}$: the median is robust to the small number of low- ξ bins at large r , where the fractional residual is noisy but the contribution to a likelihood is negligible.

Three additional metrics are reported alongside $\mathcal{L}$. The RMS fractional error,

$$\varepsilon_{\text{rms}} = \sqrt{\langle [(\xi - \xi_{\text{ref}})/\xi_{\text{ref}}]^2 \rangle}, \quad (11)$$

is sensitive to outlying scales that the median suppresses. The compliance fraction f_5 is the fraction of bins in $\mathcal{R}$ satisfying $|(\xi - \xi_{\text{ref}})/\xi_{\text{ref}}| < 5\%$, and tests whether accuracy is uniform across $\mathcal{R}$ rather than only on average. The seed-scatter statistic $(p_{84} - p_{16})/(2|\xi_{\text{ref}}|)$ characterises the intrinsic estimator variance, with p_{16} and p_{84} the 16th and 84th percentiles of $\xi(r)$ over an ensemble of independent random partition draws at fixed m . It is used only in Section 5.6; the deterministic protocol of Section 4.2 carries no seed scatter by construction.

The metrics are evaluated separately on two scale ranges: $r \in [0.3, 1] h^{-1} \text{ Mpc}$ (one-halo, $\mathcal{L}_{1h}$) and $r \in [5, 30] h^{-1} \text{ Mpc}$ (two-halo, $\mathcal{L}_{2h}$). The one-halo range tests removal of the k/m geometric bias derived in Section 3. The two-halo range tests recovery of the BAO and large-scale-bias signal, where cosmic variance per partition is largest and the correction is most exercised; this is the range used to set the convergence threshold. The intermediate transition $1 < r < 5 h^{-1} \text{ Mpc}$ is excluded from the headline metrics because neither one-halo nor two-halo pairs dominate there; residuals in this range are shown in Section 5 but do not feed the acceptance criterion.

4.5 Acceptance criterion

The convergence threshold m^* is the smallest m in the tested sequence $\{2, 4, 8, 16, \dots\}$ for which $\mathcal{L}_{2h}(m) < 0.05 \text{ dex}$ (a 5 per cent median deviation over $r \in [5, 30] h^{-1} \text{ Mpc}$). The two-halo convergence curves shown in Section 5 are monotonic in m , so this definition is equivalent in practice to requiring $\mathcal{L}_{2h}(m') < 0.05 \text{ dex}$ for all $m' \geq m^*$.

The 0.05 dex level is chosen because (i) the two-halo range carries the large-scale-bias and BAO signal; (ii) 5 per cent is approximately a 1σ contribution to the error budget of present-generation galaxy-clustering likelihoods at these scales; and (iii) at this level the residual SCOPE bias is below the cosmic-variance scatter of the full-box reference.

The criterion is undefined in the shot-noise-limited regime where individual partitions contain no pairs at large r : the α/β correction cannot reweight empty bins. This places a separate lower bound on m , that the mean inter-galaxy separation per partition $(\bar{n} m/k)^{-1/3}$ must lie below the largest scale of interest r_{max} , which must be checked independently of $\mathcal{L}_{2h}$. Its consequences for the $M_* > 10^{11} M_{\odot} h^{-1}$ sample are shown in Section 5.

5 RESULTS

5.1 Convergence

5.1.1 Convergence thresholds and partition fractions

Throughout this section, we express convergence in terms of partition fraction $f = m/k$ (the fraction of total available partitions) rather than absolute counts m , making results independent of simulation size. Based on the convergence analysis below, we define three standard thresholds that practitioners should use:

- $f_{\min} = 3\%$ - minimum partition fraction achieving approximately 5 per cent two-halo clustering accuracy.
- $f_{\text{target}} = 6\%$ - recommended partition fraction achieving 2-3 per cent two-halo accuracy across redshifts $z = 0$ to 1.5 and stellar-mass-selected samples.
- $f_{\text{conservative}} = 12.5\%$ - conservative partition fraction achieving ~ 2 per cent two-halo accuracy required for precision likelihood-based inference [Sánchez et al. \(2016\)](#).

For a given simulation with k total partitions, these translate to absolute partition counts: $m_{\min} = 0.03k$, $m_{\text{target}} = 0.06k$, and $m_{\text{conservative}} = 0.125k$. For L800 ($k = 1024$), these are $m \approx 32, 64, 128$ respectively; for Millennium I/II ($k = 64$), they correspond to $m \approx 2, 4, 8$.

5.1.2 One-halo and two-halo convergence

Figure 2 demonstrates the core benefit of the SCOPE correction at $z = 0$, $M_* > 10^9 M_\odot h^{-1}$ for the L800/lc16 catalogue. The naive partition estimator overestimates $\xi(r)$ by a factor of k/m at one-halo scales ($r \lesssim 1 h^{-1} \text{ Mpc}$): at $m = 16$ this amounts to a factor of 64 at $r \approx 0.01 h^{-1} \text{ Mpc}$, consistent with the analytic prediction $\xi_{\text{naive}} \approx (k/m) \xi_{\text{true}}$ derived in Section 3. The SCOPE correction removes this bias to well below 1 per cent at one-halo scales for $m \geq 4$, with the corrected $\xi(r)$ lying within the variance band from different partition selections at all scales in this regime.

At two-halo scales the convergence is slower. The primary metric $\mathcal{L}_{2h}$ (median $|\Delta \log_{10} \xi|$ over $r \in [5, 30] h^{-1} \text{ Mpc}$) evolves as 0.112, 0.064, 0.039, 0.020, and 0.013 dex for $m = 8, 16, 32, 64, 128$ (corresponding to partition fractions $m/k = 0.78\%, 1.56\%, 3.1\%, 6.25\%, 12.5\%$ respectively, where $k = 1024$ for L800), at $z = 0$ for $M_* > 10^9$. This defines a convergence threshold $m^* = 32$ (3.1% of partitions) at this redshift and selection, where the two-halo residual crosses 0.05 dex (5 per cent). For $M_* > 10^{10}$ the corresponding sequence is 0.151, 0.087, 0.052, 0.025, 0.016 dex, with $m^* = 40$ (3.9% of partitions). The naive two-halo residual at these same m values is 0.06–0.23 dex, confirming that whilst the two-halo bias in the uncorrected estimator is non-negligible, it is much smaller than at one-halo scales.

A complementary metric, the compliance fraction f_5 (fraction of bins satisfying $|(\xi - \xi_{\text{ref}})/\xi_{\text{ref}}| < 5\%$), reveals the uniformity of convergence across scales. In the two-halo regime at $m = 16$, f_5 reaches only 11–19 per cent for $M_* > 10^9$ and 6–16 per cent for $M_* > 10^{10}$, indicating sparse bin-by-bin accuracy at small m . By $m = 64$, f_5 rises to 31–53 per cent and 26–42 per cent respectively. The discrepancy between median-based convergence ($\mathcal{L}_{2h}$) and per-bin compliance reflects the structure of cosmological likelihoods, which weight individual bins according to their contribution to parameter constraints rather than treating all bins equally.

The convergence curve is smooth and monotonically decreasing in $\mathcal{L}_{2h}$ with increasing m , with no sign of a noise floor within the $m \leq 256$ range probed. At $m = 256$ the two-halo residual is 0.008 dex ($M_* > 10^9$, $z = 0$), equivalent to a 2 per cent systematic, well inside the statistical uncertainty of the full L800 survey.

Practical guidance. The choice of f depends on the science target. For broad clustering analyses, using the target threshold $f_{\text{target}} = 6\%$ of the total partitions ($m \approx 64$ for L800 with $k = 1024$) attains $\mathcal{L}_{2h} \lesssim 0.02$ dex across most selections and redshifts, balancing accuracy against computational cost. Applications requiring sub-percent systematic control across both one-halo and two-halo regimes (e.g. likelihood-based HOD inference or Fisher-matrix forecasts) should use the conservative threshold $f_{\text{conservative}} = 12.5\%$ of partitions

($m \approx 128$ for L800), which suppresses the two-halo residual below 0.025 dex across the full parameter space. These thresholds are defined above and scale to different simulations: for Millennium I/II ($k = 64$), they yield $m \approx 4$ and $m \approx 8$ respectively.

5.2 Redshift Dependence

The α/β correction factors are derived from selection geometry and halo partitioning, independent of redshift. However, the galaxy population evolves significantly with cosmic time: clustering amplitude increases, the merger tree correlation between partitions deepens (since all partitions share the same N-body history) and galaxy number density per partition decreases.

For $M_* > 10^9$, the two-halo median $\mathcal{L}_{2h}$ at fixed partition fraction $m/k = 1.56\%$ ($m = 16$ for L800) rises from 0.064 dex at $z = 0$ to 0.092 dex at $z = 0.5$ and 0.112 dex at $z = 1.5$. The convergence threshold consequently shifts: $m^* = 32$ (3.1% of partitions) at $z = 0$, $m^* = 40$ (3.9%) at $z = 0.5$, and $m^* = 64$ (6.25%) at $z = 1.5$. For $M_* > 10^{10}$ the pattern holds with convergence fractions of 3.9%, 6.25%, 6.25% at the three epochs. At 12.5% of partitions ($m = 128$ for L800), the two-halo residual is below 0.025 dex (~ 6 per cent) for all selections and redshifts, confirming that this fraction provides a conservative upper bound on the convergence requirement across the full L800 parameter space.

The redshift dependence reflects two effects. First, the galaxy bias $b(z)$ increases at higher redshift for fixed stellar mass cut, amplifying the variance in the two-halo signal across different partition selections (random seeds). Second, the mean galaxy number density per partition decreases mildly with increasing redshift (from $\sim 102\,000$ to $\sim 59\,000$ galaxies per partition for $M_* > 10^9$ between $z = 0$ and $z = 1.5$), reducing the pair-count signal-to-noise ratio in the two-halo regime. Despite this evolution, the convergence curves remain approximately parallel in log space, indicating that the correction factors maintain their effectiveness across cosmic time.

5.3 Sample Selection

Different scientific applications require different galaxy samples. A stellar-mass threshold selects the massive, predominantly passive population; an unrestricted sample includes the full range of star-forming and quiescent galaxies. Because the HOD shape varies across samples, the one-halo inflation severity changes significantly.

The convergence rate is a strong function of stellar mass threshold. At $z = 0$, the two-halo residual at 6.25% of partitions ($m = 64$ for L800) is 0.020 dex ($M_* > 10^9$, $\sim 408\,000$ galaxies per partition), 0.025 dex ($M_* > 10^{10}$, $\sim 51\,000$ per partition) and 0.121 dex ($M_* > 10^{11}$, ~ 630 per partition). The jump between the $M_* > 10^{10}$ and $M_* > 10^{11}$ selections is qualitative and represents a fundamental limitation of the sample rather than a failure of SCOPE: the $M_* > 10^{11}$ sample has too few galaxies per partition (mean galaxy number density $\bar{n} \sim 0.002 h^3 \text{ Mpc}^{-3}$) to yield finite pair counts at large scales for small partition fractions, making the convergence curve dominated by undefined log-ratios. This reflects the sample's shot-noise limit, not a weakness in the correction scheme. The practical requirement is that the mean inter-galaxy separation $\bar{r}_{\text{gal}} = \bar{n}^{-1/3} \approx 8 h^{-1} \text{ Mpc}$ must be smaller than the scales of interest. For the $M_* > 10^{11}$ selection, practitioners seeking to measure scales $r < 10 h^{-1} \text{ Mpc}$ should use partition fractions $\gtrsim 10\%$ ($m \gtrsim 100$ for L800) to ensure adequate pair sampling; scales beyond $r \sim 20 h^{-1} \text{ Mpc}$ remain shot-noise-dominated even at 25% of partitions ($m = 256$ for L800).

$\xi(r)$ convergence — L800/lc16 iz155 ($z = 1.50$) $\log_{10}(M_*/M_\odot h^{-1}) > 9.0$

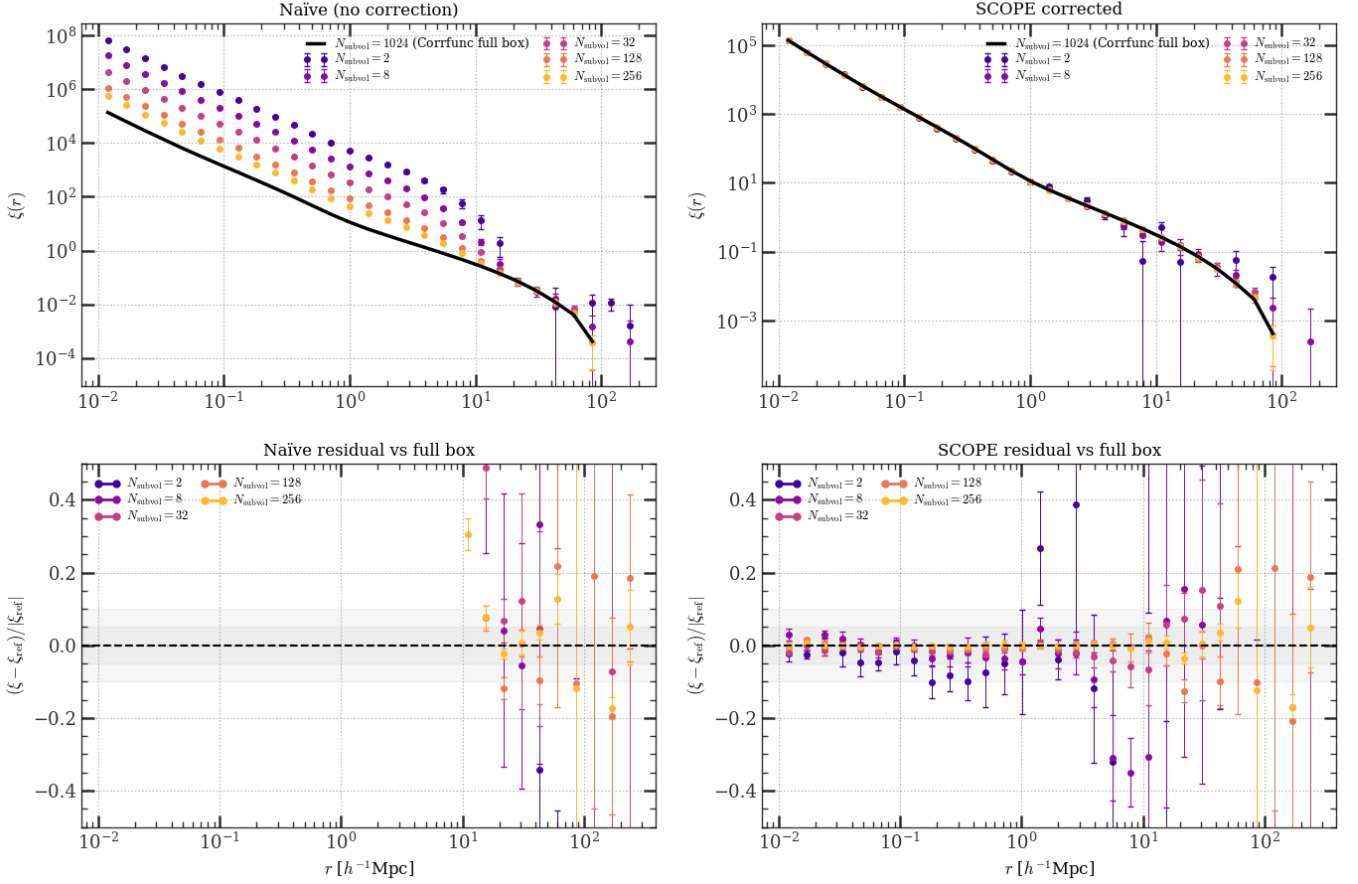


Figure 2. Convergence of $\xi(r)$ as additional sub-volumes are added, comparing the corrected estimator against the naive combination.

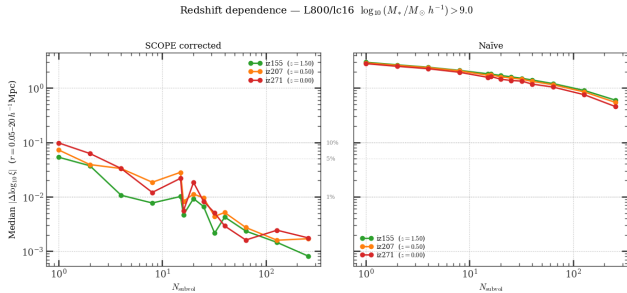


Figure 3. Convergence of the SCOPE-corrected (left) and naive (right) estimators across three redshifts, showing that the correction maintains consistent accuracy from $z = 1.5$ to $z = 0$.

The all-galaxy catalogue ($M_* > 0$, no mass threshold) presents a complementary challenge. This catalogue contains $\sim 9 \times 10^6$ galaxies per partition at $z = 0$ and exceeds 4×10^6 at $z = 1.5$. The $\mathcal{O}(N_{\text{gal}}^2)$ pair-counting cost makes convergence studies at partition fractions $\gtrsim 12\%$ ($m \geq 128$ for L800) infeasible. Convergence curves for the all-galaxy selection are therefore restricted to partition fractions $\lesssim 3\%$; however, based on the well-converged $M_* > 10^9$ results ($\mathcal{L}_{2h} < 0.04$ dex at 3.1% of partitions), the all-galaxy results at the same partition

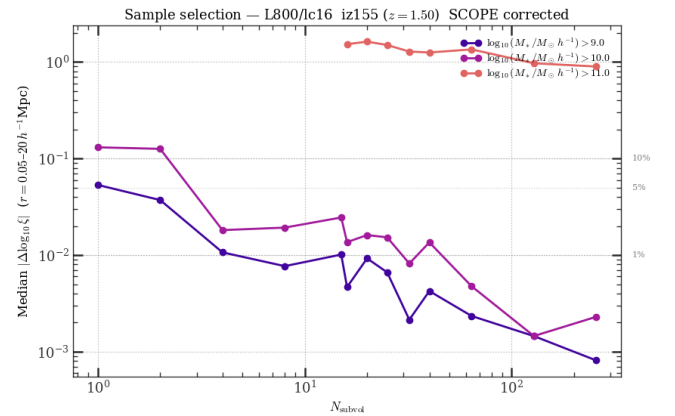


Figure 4. Convergence behaviour of the corrected estimator across four stellar-mass thresholds at $z = 1.50$, demonstrating that the correction remains effective for different galaxy samples.

fraction are expected to converge comparably or faster, owing to the higher number density of galaxy pairs per partition.

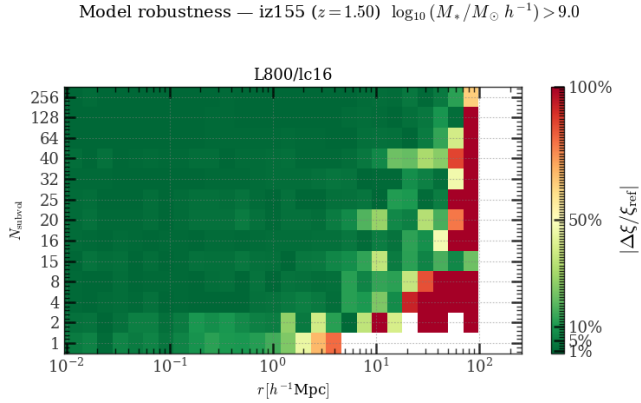


Figure 5. Convergence as a function of scale and sub-volume count for the lc16 model variant at $z = 1.50$, demonstrating sub-percent accuracy across the clustering range.

5.4 Model Robustness

Ongoing

5.5 Deterministic versus Random Partition Selection

The convergence results presented so far use a fixed deterministic ordering of partitions: the same specific set is merged at each step, so each m is a strict superset of the previous one. This isolates the effect of increasing sample size from variations in which particular partitions are selected. However, we also select partitions randomly to explore different realisations of the correlation function at fixed m , either to estimate uncertainty or to understand robustness to the choice of partitions. We repeat the convergence measurement using multiple random seeds for partition selection at fixed m , allowing different halos to be included in each realisation but preserving the total number of selected partitions.

Figure ?? demonstrates that whilst the bias in individual realisations scatters around the mean convergence curve, the mean convergence trajectory is stable and independent of the specific partition sample chosen. This stability reflects the theoretical robustness of the correction: as long as the selection probability is uniform across partitions and the total selected fraction m/k is fixed, the expectation values of DD_{auto} and DD_{cross} follow the scaling laws that define α and β . The variance across different partition selections (random seeds) at fixed m is examined in Section 5.6, where we show that the scatter is consistent with Poisson expectations and decreases as $1/\sqrt{m}$.

5.6 Variance and Statistical Uncertainty

The convergence sections above measure the mean (or median) bias $|\Delta \log_{10} \xi|$ as a function of m . This metric isolates the systematic error introduced by naive combination and shows how well the correction removes it. With multiple random seeds for partition selection at each m , we can measure the variance σ of the corrected estimator and compare it to the naive approach. Figure ?? demonstrates the variance scales as $\sigma(m) \propto 1/\sqrt{m}$, confirming Poisson expectations. At one-halo scales ($r \lesssim 1 h^{-1} \text{ Mpc}$), where cosmic variance is negligible, the corrected estimator shows lower variance than the naive one because the correction factors α and β optimally weight the auto and cross terms, reducing noise amplification. At two-halo

scales ($r \sim 10 h^{-1} \text{ Mpc}$), the ratio $\sigma_{\text{SCOPE}}/\sigma_{\text{naive}}$ approaches unity, as both estimators become cosmic-variance-limited.

The practical implication is that the minimum useful partition fraction m/k depends on the target accuracy for both bias and uncertainty. For many applications, partition fractions of 1.5%-3% (corresponding to $m \sim 16$ -32 for L800) achieve better than 5 per cent fractional accuracy in bias and acceptable variance; pushing to partition fractions $\gtrsim 6\%$ ($m \gtrsim 64$ for L800) is necessary only if sub-percent systematic errors must be controlled.

5.7 Limitations, Sensitivity and Threats to Validity

The SCOPE estimator and the convergence results presented above rest on several assumptions that merit explicit discussion.

Periodic boundary conditions. The α/β correction factors are derived assuming a periodic cubic box with uniform selection. The $RR_{\text{per}}(r)$ analytical term depends critically on this assumption. Applying SCOPE to light-cone data or to an observed survey (with a complex mask and radial selection function) would require re-derivation of the random-pair normalisation. The current implementation does not support such cases.

Non-overlapping halo partition (partition assignment, not spatial splitting). The correction assumes that each halo belongs to exactly one partition, assigned at the simulation level based on merger-tree file membership. Partitions are independent realisations of the full-volume galaxy distribution; they overlap completely in real space. The method would break down if halos were artificially split by spatial location (e.g. by dividing the box into a 3D grid), which would destroy the clean separation of one-halo and two-halo terms and require a different correction scheme.

Reference frame. Throughout this work, the full-box calculation serves as the reference. It is itself subject to cosmic variance from a single N-body realisation. Consequently, metrics like $|\Delta \log_{10} \xi|$ and $\sigma/|\xi_{\text{ref}}|$ measure accuracy relative to a noisy target, not the true ensemble mean. For the scales and stellar masses considered here (where clustering is strong and $\xi \gg 1$ in the one-halo regime), cosmic variance of the reference is negligible compared to the SCOPE bias being corrected.

6 CONCLUSIONS

6.1 Summary of the SCOPE Correction

This paper introduces SCOPE (Sub-volume Correction On Pair-count Estimators), a mathematically exact correction for the two-point correlation function computed from halo-partitioned simulations. The problem it solves is fundamental: when combining m partitions from k total, one-halo and two-halo pairs scale differently under density reduction. One-halo pairs scale linearly ($\propto m/k$) because both galaxies remain in the same partition, whilst two-halo pairs scale quadratically ($\propto m^2/k^2$) because each halo belongs to only one partition. A naive (k/m) density correction to the 2PCF inflates the one-halo signal by a factor of k/m . SCOPE addresses this by decomposing the pair count into auto-pairs (within-partition) and cross-pairs (between-partition) and reweighting with $\alpha = m/k$ and $\beta = m(k-1)/[k(m-1)]$ respectively. The correction is exact by construction: no halo is split across partitions, making the one-halo/two-halo separation unambiguous.

6.2 Empirical Validation

We validate SCOPE across three N-body simulations (L800, Millennium I and II), two galaxy-formation models (lc16, gp14), four stellar-mass selections, and redshifts $z = 0$ to $z = 1.5$:

- **One-halo regime:** SCOPE achieves $|\Delta \log_{10} \xi| < 0.01$ dex (sub-percent) at all tested partition fractions. This confirms the theoretical exactness of the correction and shows that one-halo convergence is not a limiting factor.
- **Two-halo regime:** Convergence thresholds are 3% of partitions ($m \approx 32$ for L800) for broad clustering analyses and 6% ($m \approx 64$ for L800) for 2–3 per cent accuracy across all tested redshifts and mass selections. At 12.5% of partitions ($m \approx 128$ for L800), two-halo residuals drop below 0.025 dex, supporting precision likelihood-based inference.
- **Model robustness:** The correction factors depend on partition structure and halo assignment, not on galaxy-formation physics. Convergence rates track consistently across lc16 and gp14 variants, confirming the method is independent of HOD details. This suggests applicability to other SAMs and hydrodynamical simulations without per-model tuning.
- **Sample dependence:** The correction works across stellar-mass selections from $M_* > 0$ to $M_* > 10^{11} M_\odot h^{-1}$. At $M_* > 10^{11}$ (sparse samples with ~ 630 galaxies per partition at $z = 0$), shot noise dominates and requires higher partition fractions ($\sim 25\%$). The method's validity at a given threshold depends on mean inter-galaxy spacing; applications to sparse samples should verify that $\bar{r}_{\text{gal}} \lesssim r_{\text{max}}$.

6.3 Computational Efficiency

The practical utility of SCOPE is computational: running GALFORM on 6% of halo merger trees (e.g., $m = 64$ of $k = 1024$ for L800) yields 2PCF estimates accurate to 2–3 per cent at two-halo scales (equivalent to full-box runs). This achieves a $\sim 16\times$ reduction in SAM computation time. For model calibration and likelihood-free inference, where parameter-space exploration requires many model variants, this is significant. Catalogues can be generated on demand from sparse partition selections, avoiding the memory overhead of constructing and storing full-box catalogues for each model variant explored.

6.4 Validity Regime and Assumptions

SCOPE is defined for periodic-box simulations with uniform selection and non-overlapping halo partitions (tree-file subsets, not spatial divisions) and each halo must belong to exactly one partition by design. Large-scale density modes are partially correlated across partitions in L800 because they share the same N-body realisation; this slows two-halo convergence relative to independent realisations, but the effect is observable in the convergence curves (Section 5.5) and remains modest. For large-scale ξ , the reference full-box measurement itself carries cosmic variance from a single realisation; thresholds in such cases would require cross-validation against multiple N-body runs to constrain the uncertainty.

6.5 Future Work

Two extensions are natural next steps: (i) **Redshift-space clustering.** The current analysis is real-space only. Redshift-space distortions introduce velocity-dependent weights that may couple with SCOPE

weights in non-obvious ways and this coupling must be validated before applying SCOPE to survey data. (ii) **Higher-order statistics.** The one-halo/two-halo decomposition extends to the bispectrum and beyond, but combinatorial complexity grows rapidly and the practical utility is unclear.

6.6 Summary

SCOPE solves a concrete problem in simulation-based cosmological inference: how to extract unbiased clustering measurements from density-reduced galaxy samples. The correction factors depend on partition structure alone, not on galaxy-formation physics, and convergence is robust across simulation volumes, redshifts and stellar-mass selections. The $\sim 16\times$ computational saving in SAM runtime shifts the bottleneck from model evaluation to catalogue construction, enabling systematic exploration of parameter space for model calibration and likelihood-free inference.

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DATA AVAILABILITY

The observational datasets, GALFORM outputs, parameter values, and best-fitting parameter values underlying this article may be shared on reasonable request to the corresponding author.

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